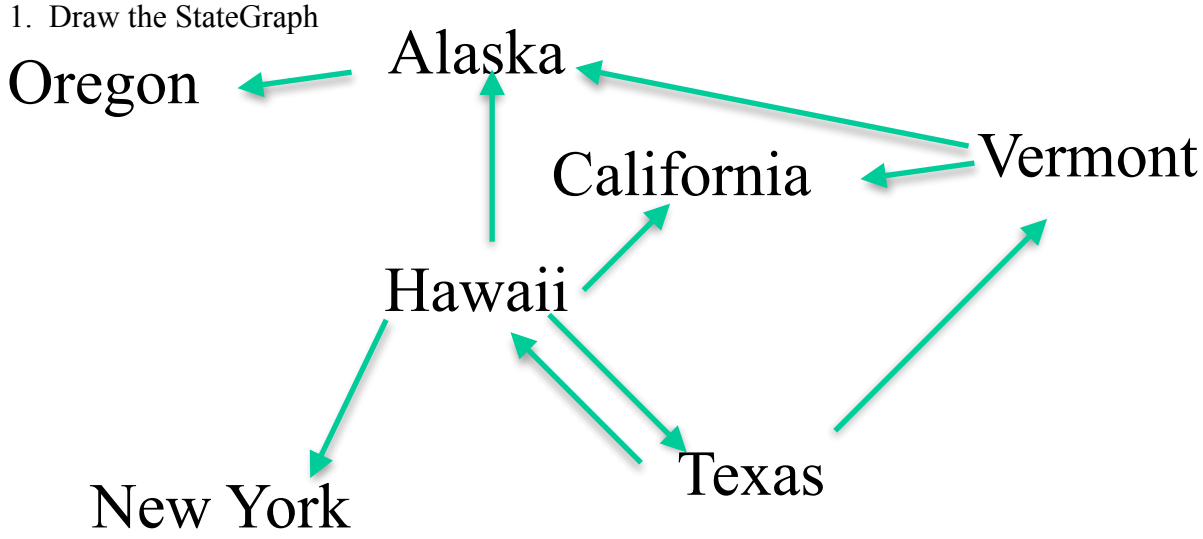
CMSC204
Kartchner

V(StateGraph) = {Oregon, Alaska, Texas, Hawaii, Vermont, NewYork, California}
E(StateGraph) = {(Alaska, Oregon), (Hawaii, Alaska), (Hawaii, Texas), (Texas, Hawaii), (Hawaii, California), (Hawaii, New York), (Texas, Vermont), (Vermont, California), (Vermont, Alaska)}

1. Draw the StateGraph



1. Describe the graph pictured above, using the formal graph notation.

V(StateGraph) = **{Oregon, Alaska, Texas, Hawaii, Vermont, NewYork, California}**

E(StateGraph) = **{(Alaska, Oregon), (Hawaii, Alaska), (Hawaii, Texas), (Texas, Hawaii), (Hawaii, California), (Hawaii, New York), (Texas, Vermont), (Vermont, California), (Vermont, Alaska)}**

2. a. Is there a path from Oregon to any other state in the graph?

No, there is no path from Oregon to any other state.

b. Is there a path from Hawaii to every other state in the graph?

Yes, there is a path from Hawaii to every other state.

c. From which state(s) in the graph is there a path to Hawaii?

Only Texas has a direct path to Hawaii.

3. a. Show the adjacency matrix that would describe the edges in the graph. Store the vertices in alphabetical order

States

Alaska

California

Hawaii

New York

Oregon

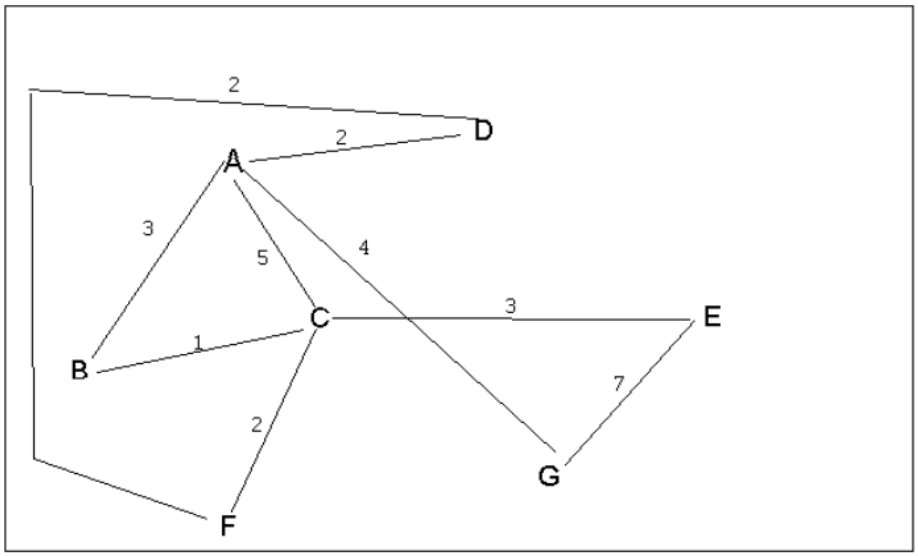
Texas

Vermont

	AK	CA	HI	NY	OR	TX	VT
AK	0	0	0	0	1	0	0
CA	0	0	0	0	0	0	0
HI	1	1	0	1	0	1	0
NY	0	0	0	0	0	0	0
OR	0	0	0	0	0	0	0
TX	0	0	1	0	0	0	1
VT	1	1	0	0	0	0	0

3. b. Show the adjacency lists that would describe the edges in the graph

Alaska - {Oregon}
California - {}
Hawaii - {Alaska, California, New York, Texas}
New York - {}
Oregon - {}
Texas - {Hawaii, Vermont}
Vermont - {Alaska, California}

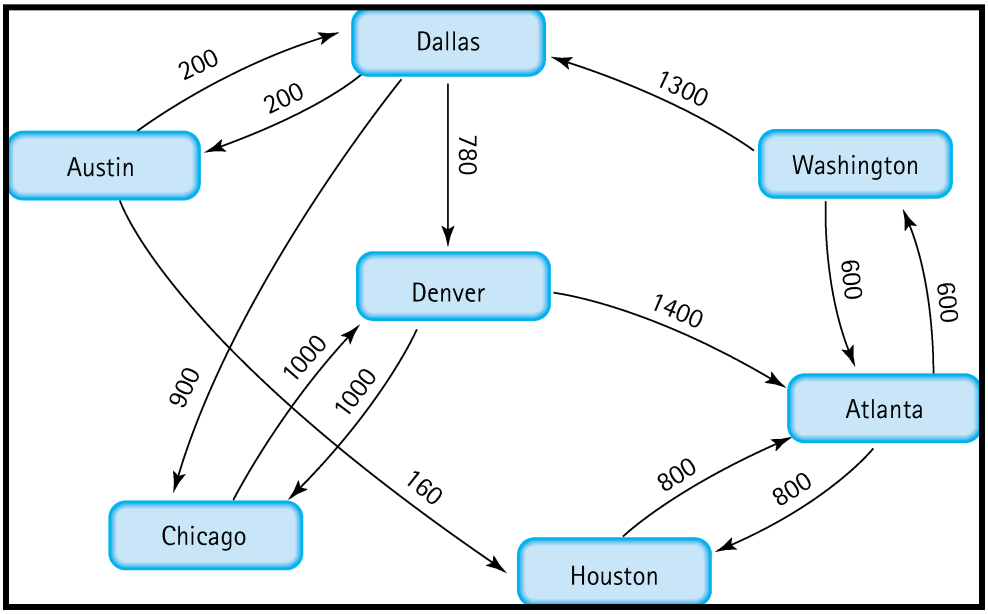


4 a. Which of the following lists the graph nodes in depth first order beginning with E?

A) E, G, F, C, D, B, A
B) G, A, E, C, B, F, D
C) E, G, A, D, F, C, B
D) E, C, F, B, A, D, G

4 b. Which of the following lists the graph nodes in breadth first order beginning at F?

A) F, C, D, A, B, E, G
B) F, D, C, A, B, C, G
C) F, C, D, B, G, A, E
D) a, b, and c are all breadth first traversals



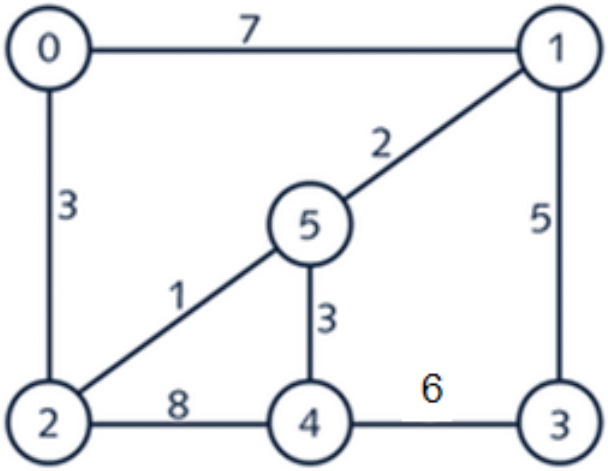
5. Find the shortest <u>distance</u> from Atlanta to every other city

Distances: Atlanta = 0 Washington = ∞ Houston = ∞ Dallas = ∞ Austin = ∞ Denver = ∞ Chicago = ∞

Step 1. Start at Atlanta (0)
Updated distances:
Washington = 600
Houston = 800
Step 2. Visit Washington (600)
Updated distances:
Dallas = 1900
Step 3. Visit Houston (800) no updates
Step 4. Visit Dallas (1900)
Updated distances:
Chicago = 2800
Denver = 2680
Step 5. Visit Austin(2100) no updates
Step 6. Visit Denver (2680) no updates
Step 7. Visit Chicago (2800) no updates

CITY. DISTANCE. PATH TAKEN.
Atlanta 0 —
Washington 600 Atlanta - Washington
Houston 800 Atlanta - Houston
Dallas 1900 Atlanta - Washington - Dallas
Austin 2100 Atlanta - Washington - Dallas - Austin
Denver 2680 Atlanta - Washington - Dallas - Denver
Chicago 2800 Atlanta - Washington - Dallas - Chicago

6. Find the minimal spanning tree using Prim's algorithm. Use 0 as the source vertex . Show the steps.



Step 0 – Initialization

- Start at vertex: **0**
- MST = { }
- Visited = {0}

Step 1 – Add edge (0, 2), weight = 3

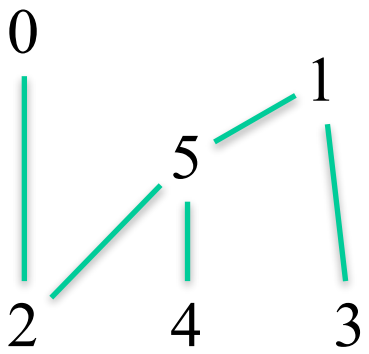
- Add edge (0, 2) to MST
- Visited = {0, 2}

Step 2 – Add edge (2, 5), weight = 1

- Add edge (2, 5) to MST
- Visited = {0, 2, 5}

Step 3 – Add edge (5, 1), weight = 2

- Add edge (5, 1) to MST
- Visited = {0, 1, 2, 5}



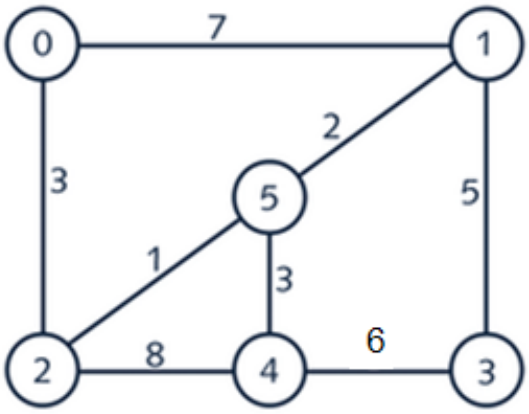
Step 4 – Add edge (1, 3), weight = 5

- Add edge (1, 3) to MST
- Visited = {0, 1, 2, 4, 5}

Step 5 – Add edge (5 4), weight = 3

- Add edge (5, 4) to MST
- Visited = {0, 1, 2, 3, 4, 5}

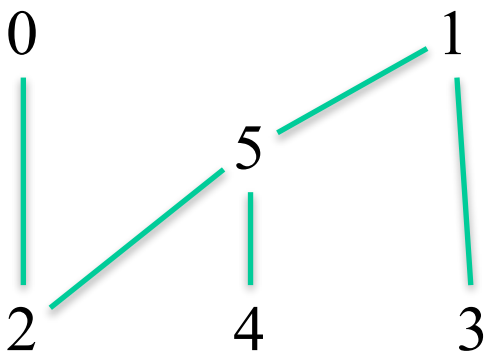
7. Find the minimal spanning tree using Kruskal's algorithm. Show the weights in order and the steps.



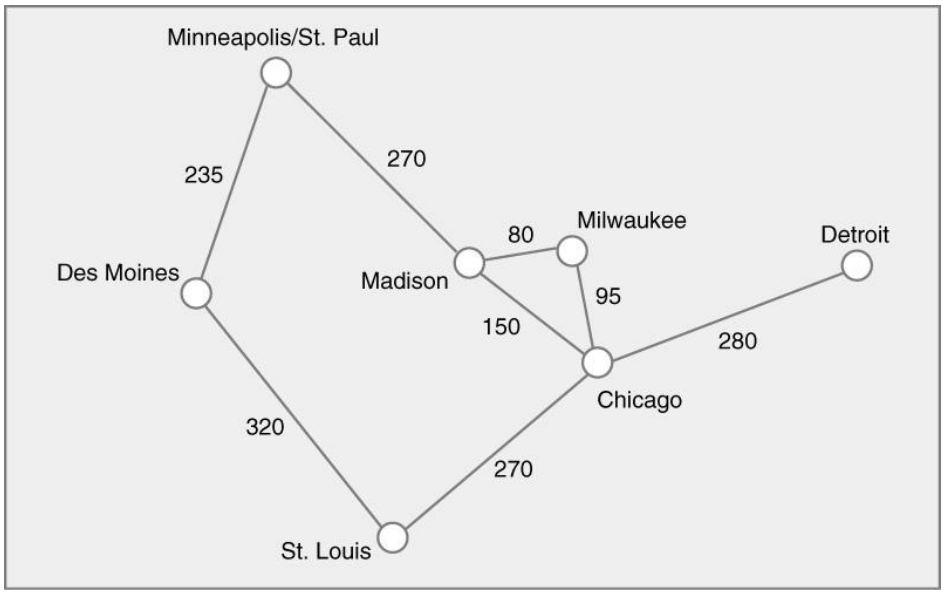
1. Add **(2, 5)** — weight 1
   - MST: {(2, 5)}
2. Add **(5, 1)** — weight 2
   - MST: {(2, 5), (5, 1)}
3. Add **(0, 2)** — weight 3
   - MST: {(2, 5), (5, 1), (0, 2)}
4. Add **(5, 4)** — weight 3
   - MST: {(2, 5), (5, 1), (0, 2), (5, 4)}
5. Add **(1, 3)** — weight 5
   - MST: {(2, 5), (5, 1), (0, 2), (5, 4), (1, 3)}
   - Connected: All vertices (0 to 5)

All remaining edges (4, 3), (0, 1), (2, 4) form cycles

Weight = 1 + 2 + 3 + 3 + 5 = 14



8. Find the minimal spanning tree using the algorithm you prefer. Use Minneapolis/St. Paul as the source vertex



Step 1 Start at **Minneapolis:** Choose (Minneapolis, **Des Moines**) = 235
Visited: Minneapolis, Des Moines

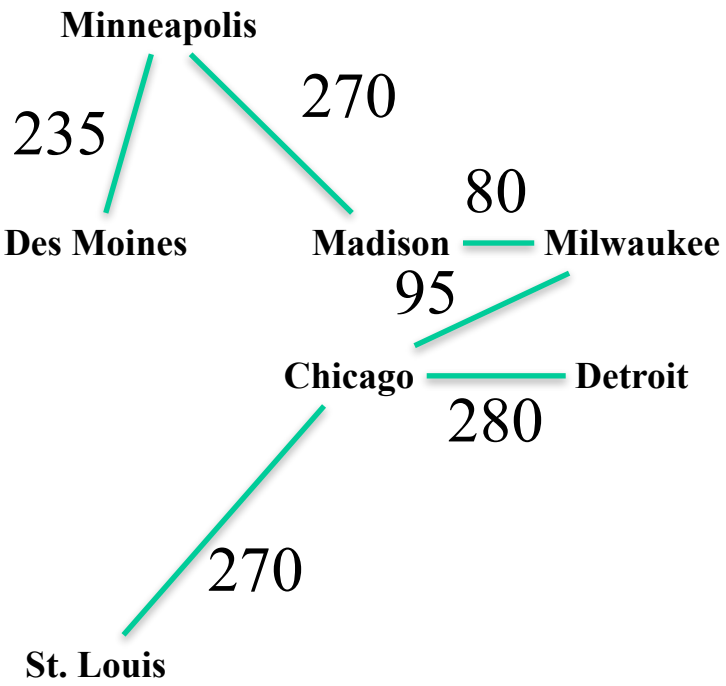
Step 2: Next lowest: (Minneapolis, **Madison**) = 270
Visited: + Madison

Step 3 Madison: Pick (Madison, **Milwaukee**) = 80
Visited: + Milwaukee

Step 4 Milwaukee: Pick (Milwaukee, **Chicago**) = 95
Visited: + Chicago

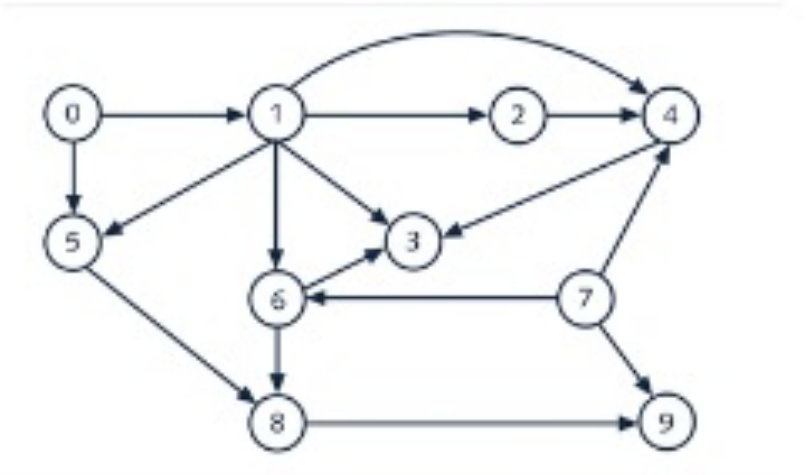
Step 5 Chicago: Pick (Chicago, **St. Louis**) = 270
Visited: + St. Louis

Step 6 Chicago: Pick (Chicago, **Detroit**) = 280
Visited: + Detroit



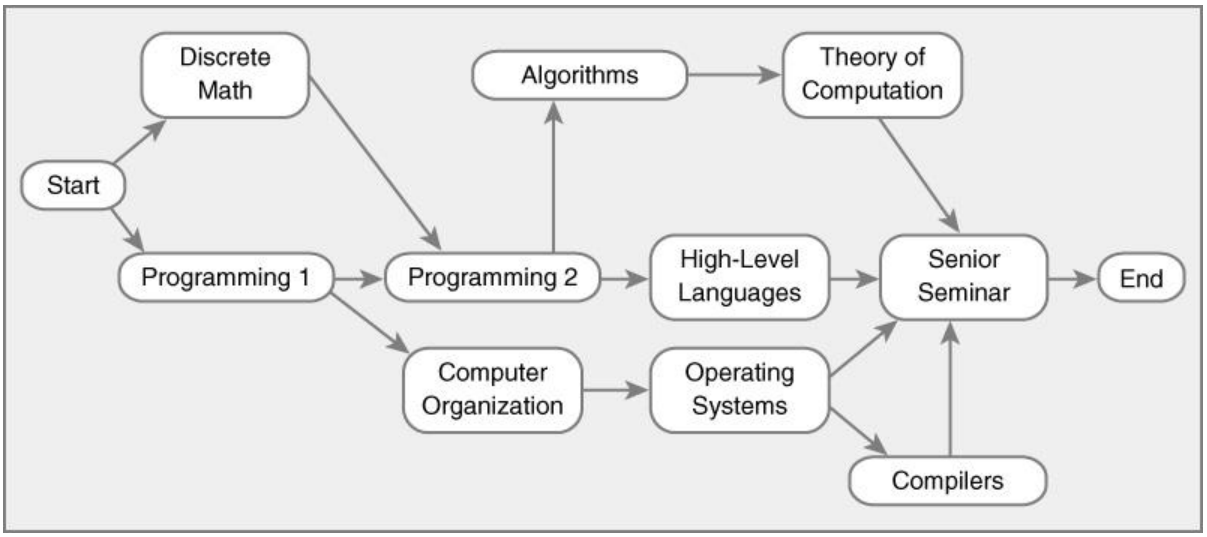
9. List the nodes of the graph in a breadth first topological ordering. Show the steps using arrays predCount, topologicalOrder and a queue

For simplicity and convince topologicalOrder was reduced to Order*



Step 1
Add 0 to topologicalOrder, Decrease predCount of 1 (0), 5 (0) ,Queue: [1, 5] ,Order:[0]
Step 2
Decrease predCount of 2 (0), 3 (1), 4 (1) ,Queue: [5, 2], Order: [0, 1]
Step 3
Decrease predCount of 6 (2), 8 (2), Queue: [2], Order: [0, 1, 5]
Step 4
Decrease predCount of 4 (0), Queue: [4], Order: [0, 1, 5, 2]
Step 5
Decrease predCount of 6 (1), 7 (0), Queue: [7],Order: [0, 1, 5, 2, 4]
Step 6
Decrease predCount of 8 (1), 9 (1), Queue: [],Order: [0, 1, 5, 2, 4, 7]
Step 7
Decrease predCount of 6 (0),Queue: [6],Order: [0, 1, 5, 2, 4, 7, 3]
Step 8
Decrease predCount of 8 (0),Queue: [8],Order: [0, 1, 5, 2, 4, 7, 3, 6]
Step 9
Decrease predCount of 9 (0),Queue: [9],Order: [0, 1, 5, 2, 4, 7, 3, 6, 8]
FINAL
Queue is empty
Final Order: [0, 1, 5, 2, 4, 7, 3, 6, 8, 9]

10. List the nodes of the graph in a breadth first topological ordering.



Final Answer:

[Start, Programming 1, Discrete Math, Computer Organization, Programming 2, Operating Systems, Algorithms, High-Level Languages, Compilers, Theory of Computation, Senior Seminar, End]